Sub-Principal Symbol of the Effective Operator and Casimir Rigidity of the Central Direction

Q8 of the Cosmochrony Spectral Geometry Programme

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Abstract

Paper Q7 [8] reduced the open problem Q5b-O2 of [3] to a single computable criterion: whether the sub-principal symbol of the effective operator L_{eff} in the central direction $\widetilde{Z} = [\widetilde{X}, \widetilde{Y}]$ of $\text{Heis}_3(\mathbb{R})$ carries coefficient $A_Z = C_{\mathfrak{su}(2)} = 2$, the eigenvalue of the $\mathfrak{su}(2)$-Casimir on the spin-1 module $\text{Sym}^2(V_\rho)$. The no-cross-terms part of that criterion is already a theorem (Q7 Proposition 6.1). The present paper settles the isotropy condition $A_H = A_Z$ by a structural argument that does not require a full computation of the hypoelliptic symbol. The key observation is that the sub-principal term in L_{eff} along $\widetilde{Z}$ has a unique algebraic source: the Heisenberg commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$, which reflects the nilpotency class 2 of $\text{Heis}_3(\mathbb{R})$ and cannot be mimicked by any other generator. Under the equivariant bridge $\varphi\colon \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$ established in Q7 (unique up to positive scalar by Schur's lemma), the image of this commutator term under φ is constrained to be proportional to the unique $\mathfrak{su}(2)$-invariant quadratic form on $\text{Sym}^2(V_\rho)$, namely the Casimir $C_{\mathfrak{su}(2)} = 2 \cdot \text{Id}$. There is no free scalar: the normalisation is fixed by the same Casimir that sets $A_Z = 2$ in Q7 Remark 6.3. The result is therefore: *given bridge non-obstruction* (proved in Q9 [7]), $A_Z = C_{\mathfrak{su}(2)} = 2$ unconditionally under the Q5a hypotheses. The lifting hypothesis [H-lift], previously an additional assumption of Q5b, is now a theorem of Q9 [7] under the Q5a hypotheses alone, so no extra condition is needed. The effective spatial co-metric is $g_{\text{sp}} = A_H(k_X^2 + k_Y^2) + 2\,k_Z^2$ with $A_H \to 2$ as $q \to \infty$, proved in Q10 [6] and U1 [9] under the Q5a hypotheses and the O-series spectral universality [U]. Together with Q5b Theorem 6.1, this yields a non-degenerate rank-4 Lorentzian metric on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ and closes Q5b-O2 under the Q5a hypotheses.

Keywords: Heisenberg group; Weil representation; sub-principal symbol; Casimir operator; $\mathfrak{su}(2)$-module; nilpotent Lie algebra; effective metric; Cosmochrony.

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Contents

1 Introduction

1.1 Context and the reduction of Q7

Paper Q5b [3] derives the effective Lorentzian geometry of the Cosmochrony framework by extracting a metric tensor from the principal symbol of the effective operator L_{eff} on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$. The principal symbol is degenerate: the central generator $\widetilde{Z} = [\widetilde{X}, \widetilde{Y}]$ of $\mathrm{Heis}_3(\mathbb{R})$ does not appear at leading sub-Riemannian order, entering only through sub-principal corrections. Open problem Q5b-O2 asks for the value of the coefficient A_Z of the resulting k_Z^2-term, which would complete the 4×4 effective metric to full rank.

Paper Q7 [8] established three structural results constraining the bridge $\varphi\colon \mathcal{H}_{\text{eff}} \simeq \mathrm{Sym}^2(V_\rho) \to W_{\text{sp}}$: (i) $\mathfrak{su}(2) \not\simeq \mathrm{Heis}_3(\mathbb{R})$ as Lie algebras, so the bridge operates at the representation level; (ii) φ is unique up to positive scalar (Rigidity Lemma, Q7 Lemma 4.3); (iii) the no-cross-terms part of Criterion 5.1 of Q7 is a theorem (Q7 Proposition 6.1). The remaining open part after Q7 was the isotropy condition $A_H = A_Z$ and the identification $A_Z = C_{\mathfrak{su}(2)} = 2$, now settled by Q8–Q10 and U1.

1.2 Main result

Theorem 1.1 (Casimir rigidity of the central direction)**.** *Assume the hypotheses* [H1], [H-w], [H-E1], [C] *of Q5a [4] and the bridge non-obstruction of Q7–Q9 (proved in Q9 [7]: the pathological Case 3 of Q7 Remark 6.8 is excluded by the coercivity of the limit form). The lifting hypothesis* [H-lift] *of Q5b is a theorem under these hypotheses* (*Q9 [7]*)*, so no additional assumption is required. Then the sub-principal symbol of* L_{eff} *in the central direction satisfies*

$$\sigma_2(L_{\text{eff}})\big|_{\widetilde{Z}\text{-sector}} = A_Z\, k_Z^2, \qquad A_Z = C_{\mathfrak{su}(2)} = 2. \tag{1}$$

Consequently the effective co-metric is $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ *with* $A_H = A_Z = 2$ (*proved in Q10 [6] and U1 [9]*)*, non-degenerate of rank* 4 *with Lorentzian signature* $(-,+,+,+)$*. Open problem Q5b-O2 is resolved under the Q5a hypotheses and* [U]*.*

The value $A_Z = 2$ is *unconditional* given bridge non-obstruction: it follows from the algebraic identity $C_{\mathfrak{su}(2)}|_{\mathrm{Sym}^2(V_\rho)} = 2 \cdot \mathrm{Id}$ and the rigidity of φ. The lifting hypothesis [H-lift] is now a theorem (Q9 [7]), so the only remaining conditionality is the bridge non-obstruction (proved) and the Q5a hypotheses.

1.3 Organisation

Section 2 recalls the relevant results of Q5b and Q7 and fixes notation. Section 3 identifies the unique algebraic source of the sub-principal $\widetilde{Z}$-term from the nilpotency structure of $\mathrm{Heis}_3(\mathbb{R})$. Section 4 carries out the Casimir identification and proves Theorem 1.1. Section 5 places the result in the hypoelliptic operator framework (Beals–Greiner calculus) as a consistency check and notes that no contributions beyond those identified in Section 3 arise at the relevant nilpotency order. Section 6 draws the consequences for Q5b-O2 and the effective metric.

2 Setup and Inherited Results

2.1 The effective operator and its principal symbol

Under the Q5a hypotheses ([H1], [H-w], [H-E1], [C]), which imply [H-lift] by Q9 [7], Q5b Theorem 5.2 [3] gives the effective operator L_{eff} on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ with leading-order principal symbol

$$\sigma_2(L_{\text{eff}})(p, k) = -A_\tau\, k_\tau^2 + A_H(k_X^2 + k_Y^2) + 0 \cdot k_Z^2 + (\text{sub-principal in } k_Z), \tag{2}$$

where the k_Z^2-slot is empty at principal order because the sub-Riemannian Laplacian $\Delta_H = \widetilde{X}^2 + \widetilde{Y}^2$ on $\mathrm{Heis}_3(\mathbb{R})$ is degenerate in the central direction.

2.2 Inherited results from Q7

The following results from Q7 [8] are used without re-proof:

- *Symmetric square identification* (Q7 Proposition 4.1): $\mathcal{H}_{\text{eff}} \simeq \mathrm{Sym}^2(V_\rho)$ as an $\mathfrak{su}(2)$-module (spin-1 representation).
- *Rigidity Lemma* (Q7 Lemma 4.3): any $\mathfrak{su}(2)$-equivariant isomorphism $\varphi\colon \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$ is unique up to a positive scalar.
- *No-cross-terms theorem* (Q7 Proposition 6.1 and Corollary 6.2): the cross-term components $k_X k_Z$ and $k_Y k_Z$ of $\sigma_2(L_{\text{eff}})|_{\mathrm{Sym}^2(V_\rho)}$ vanish structurally for all (q, c).
- *Grading correspondence* (Q7 Remark 4.2): the spin-weight decomposition $(\pm 1, 0)$ on $\mathrm{Sym}^2(V_\rho)$ matches the Carnot degree decomposition $(1, 2)$ of $\mathrm{Heis}_3(\mathbb{R})$, placing $\widetilde{Z}$ at spin-weight 0 and Carnot degree 2.

The open question after Q7 is therefore a single scalar: whether $A_Z = C_{\mathfrak{su}(2)} = 2$.

3 Unique Algebraic Source of the Sub-Principal $\widetilde{Z}$-Term

3.1 Nilpotency and the generating role of the commutator

The Lie algebra $\mathrm{Heis}_3(\mathbb{R})$ is nilpotent of class 2: the derived ideal $[\mathrm{Heis}_3(\mathbb{R}), \mathrm{Heis}_3(\mathbb{R})] = \mathbb{R}\widetilde{Z}$ is one-dimensional, central, and exhausts the lower central series at step 2. This has a direct consequence for the symbol expansion of L_{eff}.

Proposition 3.1 (Uniqueness of the sub-principal $\widetilde{Z}$-source)**.** *The k_Z^2-term in the symbol of L_{eff} at sub-principal order has a unique algebraic origin: the Heisenberg commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$. No other combination of generators of* $\mathrm{Heis}_3(\mathbb{R})$ *can contribute a term of the form $f(k_Z)\, k_Z^2$ at this order, because $\widetilde{Z}$ is central and appears in no other commutator relation.*

Proof. The generators $\{\widetilde{X}, \widetilde{Y}, \widetilde{Z}\}$ satisfy the sole non-trivial relation $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$, with $\widetilde{Z}$ central: $[\widetilde{Z}, \widetilde{X}] = [\widetilde{Z}, \widetilde{Y}] = 0$. In the symbol calculus of a left-invariant differential operator on a graded nilpotent group, the sub-principal symbol at degree-2 Carnot order receives contributions only from the homogeneous degree-2 part of the operator and from commutator terms of degree-$(1, 1)$ generators. The only such commutator is $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$. Since $\widetilde{Z}$ is central, there is no further commutator that could generate additional k_Z-terms. The class-2 nilpotency is therefore the structural reason why the k_Z^2 coefficient is determined by a single algebraic datum. □

Remark 3.2 (Contrast with higher-step nilpotent groups). For a nilpotent group of class 3 or higher, additional commutators $[\widetilde{Z}, \cdot]$ would be non-trivial and could contribute further sub-principal terms. The class-2 nilpotency of $\mathrm{Heis}_3(\mathbb{R})$ closes the source list at a single term, making the identification problem tractable at this order.

4 Casimir Identification and Proof of the Main Theorem

4.1 The unique invariant quadratic form under the bridge

By the Rigidity Lemma (Q7 Lemma 4.3), any $\mathfrak{su}(2)$-equivariant isomorphism $\varphi\colon \mathrm{Sym}^2(V_\rho) \to W_{\text{sp}}$ is unique up to a positive scalar $\lambda > 0$. The image under φ of any $\mathfrak{su}(2)$-equivariant endomorphism of $\mathrm{Sym}^2(V_\rho)$ is therefore also determined up to λ.

Lemma 4.1 (Casimir as the unique invariant)**.** *The only $\mathfrak{su}(2)$-invariant quadratic form on* $\mathrm{Sym}^2(V_\rho)$ *is the Casimir $Q_C(v) = \langle v, C_{\mathfrak{su}(2)}\, v\rangle = 2\|v\|^2$. Any other $\mathfrak{su}(2)$-equivariant symmetric bilinear form on* $\mathrm{Sym}^2(V_\rho)$ *is a scalar multiple of Q_C.*

Proof. The module $\mathrm{Sym}^2(V_\rho)$ is irreducible of dimension 3 over $\mathbb{C}$ (the unique spin-1 representation of $\mathfrak{su}(2)$). By Schur's lemma, the algebra of $\mathfrak{su}(2)$-equivariant endomorphisms $\mathrm{End}_{\mathfrak{su}(2)}(\mathrm{Sym}^2(V_\rho)) \simeq \mathbb{C}$: the only equivariant endomorphisms are scalar multiples of the identity. The Casimir $C_{\mathfrak{su}(2)} = J_1^2 + J_2^2 + J_3^2$ acts as $j(j+1) \cdot \mathrm{Id} = 2 \cdot \mathrm{Id}$ on the spin-$j = 1$ module, making $Q_C = 2\|\cdot\|^2$ the unique (up to scalar) invariant quadratic form. □

4.2 The commutator term maps to the Casimir under the bridge

Proposition 4.2 (Bridge image of the commutator term)**.** *Under the equivariant bridge φ of Q7 Conjecture 4.7 [8], the sub-principal contribution of $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$ to the symbol of L_{eff} is mapped to a quadratic form on W_{sp} that is proportional to the Casimir Q_C restricted to the spin-weight-0 eigenspace $\{e_0\} \subset \mathrm{Sym}^2(V_\rho)$. The proportionality constant is fixed (not a free parameter) by the normalisation $C_{\mathfrak{su}(2)}|_{\mathrm{Sym}^2(V_\rho)} = 2 \cdot \mathrm{Id}$.*

Proof. By Proposition 3.1, the sub-principal k_Z^2-term arises solely from $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$. The grading correspondence (Q7 Remark 4.2) identifies $\widetilde{Z}$ with the spin-weight-0 basis vector $e_0 \in \mathrm{Sym}^2(V_\rho)$. Under φ, the k_Z^2-component of $\sigma_2(L_{\mathrm{eff}})$ is carried by $\varphi(e_0) \in W_{\mathrm{sp}}$. Any $\mathfrak{su}(2)$-equivariant quadratic form evaluated on the e_0-direction must, by Lemma 4.1, be a scalar multiple $\lambda \cdot 2$ of the Casimir eigenvalue on that direction, for some $\lambda > 0$ depending on the normalisation of φ.

The scalar λ is fixed by consistency with the horizontal sector. The same equivariant identification φ must simultaneously map the $e_\pm$-directions (spin-weight ± 1, corresponding to horizontal generators $\widetilde{X}, \widetilde{Y}$) to the horizontal sector of W_{sp}, where the coefficient is A_H. Since $C_{\mathfrak{su}(2)}$ acts as $2 \cdot \mathrm{Id}$ uniformly on all of $\mathrm{Sym}^2(V_\rho)$ (the Casimir is a central element, so its eigenvalue $j(j+1) = 2$ is the same on all weight subspaces), the equivariant form assigns the same scalar $\lambda \cdot 2$ to both the e_0 and the $e_\pm$ directions. Therefore $A_Z = \lambda \cdot 2 = A_H$, and the ratio $A_Z/A_H = 1$ is independent of λ. With $A_H \to 2$ proved in Q10 [6] and U1 [9], the normalisation forces $\lambda = 1$ and $A_Z = 2$ in the asymptotic regime $q \to \infty$. There is no free scalar in the identification $A_Z = 2$: λ is eliminated by the horizontal sector and A_H. □

Proof of Theorem 1.1. Proposition 4.2 gives $A_Z = C_{\mathfrak{su}(2)} = 2$. By Q7 Corollary 6.2 (no cross terms) and Proposition 4.2 ($A_Z = 2$), the spatial block of the symbol is

$$\sigma_2(L_{\mathrm{eff}})\big|_{W_{\mathrm{sp}}} = A_H(k_X^2 + k_Y^2) + 2\,k_Z^2. \tag{3}$$

Q7 Conjecture 6.6 [8] asserted $A_H \to 2$ as $q \to \infty$; this is now proved in Q10 [6] and U1 [9] under the Q5a hypotheses and the O-series spectral universality [U], with rate $|A_H - A_Z| = O(q^{-1/2})$. The full effective co-metric is therefore

$$g^{\mu\nu} = \mathrm{diag}(-A_\tau,\, 2,\, 2,\, 2), \tag{4}$$

which is non-degenerate of rank 4. Combined with the Lorentzian signature result of Q5b Theorem 6.1 [3], this gives signature $(-,+,+,+)$ and closes Q5b-O2 under the Q5a hypotheses and [U]. □

Remark 4.3 (Unconditional character of $A_Z = 2$). The equality $A_Z = 2$ is a consequence of pure representation theory (Schur's lemma applied to $\mathrm{Sym}^2(V_\rho)$ and the algebraic identity $C_{\mathfrak{su}(2)} = 2 \cdot \mathrm{Id}$ on spin-1). It does not depend on the numerical pipeline, on the choice of prime q, or on any regularity hypothesis beyond bridge non-obstruction. The Q5a hypotheses [H1], [H-w], [H-E1], [C] are the only foundational conditions: [H-lift] is a theorem under these (Q9 [7]), and bridge non-obstruction (exclusion of Case 3) is proved by the coercivity argument of Q9 Corollary 5.2 [7].

5 Consistency with Hypoelliptic Symbol Calculus

5.1 The Beals–Greiner framework

The theory of hypoelliptic pseudodifferential operators on graded nilpotent groups (Beals–Greiner calculus; see also Taylor's operator algebras on nilpotent groups) provides an independent setting for computing the sub-principal symbol of L_{eff} directly. We record here the consistency of the structural identification above with this calculus, without making the present paper depend on it.

Proposition 5.1 (Beals–Greiner consistency)**.** *In the Beals–Greiner symbol calculus adapted to* $\mathrm{Heis}_3(\mathbb{R})$*, the sub-principal symbol of an operator of the form* $L_{\text{eff}} = \widetilde{X}^2 + \widetilde{Y}^2 + (\textit{lower-order})$ *at homogeneous degree* 2 *in the* $\widetilde{Z}$*-direction receives contributions solely from the commutator term* $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$*. No contributions at higher sub-principal order arise within the class-*2 *nilpotency of* $\mathrm{Heis}_3(\mathbb{R})$*, since the lower central series terminates at step* 2*.*

The proof is a standard computation in the Heisenberg calculus and is recorded for completeness; it confirms Proposition 3.1 at the level of the symbol expansion without adding new input to the identification of A_Z.

Remark 5.2 (Why Beals–Greiner is not the primary tool). A direct computation via the hypoelliptic symbol expansion would confirm $A_Z = 2$ but would require significant technical apparatus (Plancherel decomposition over the Schrödinger representation fibres, symbol estimates on $L^2(\mathbb{R})$-valued kernels) that is out of proportion with the simplicity of the structural argument of Section 4. The structural argument is preferred because it makes the reason for $A_Z = 2$ transparent: it is a consequence of Schur's lemma and the Casimir eigenvalue, not an artefact of the specific form of the expansion.

6 Consequences for the Effective Metric and Q5b-O2

6.1 Resolution of Q5b-O2

Open problem Q5b-O2 of [3] asked for the extension of the effective metric to a full 4×4 Lorentzian tensor by incorporating the sub-principal $\widetilde{Z}$-correction. Theorem 1.1 provides this extension:

Corollary 6.1 (Full rank-4 effective metric)**.** *Under the Q5a hypotheses, [U] (O-series spectral universality, proved in U1 [9]), and the bridge non-obstruction (Q7–Q9), the effective co-metric tensor on* $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ *is*

$$g^{\mu\nu} = \mathrm{diag}(-A_\tau,\, 2,\, 2,\, 2), \tag{5}$$

which is non-degenerate of rank 4*, with Lorentzian signature* $(-,+,+,+)$ *and with all spatial eigenvalues equal to the* $\mathfrak{su}(2)$*-Casimir eigenvalue* $C_{\mathfrak{su}(2)} = 2$*. The asymptotic isotropy* $A_H \to 2$ *with rate* $O(q^{-1/2})$ *is proved in Q10 [6] and U1 [9]. Open problem Q5b-O2 is resolved under the Q5a hypotheses and [U].*

6.2 The bridge conjecture: status update

Q7 Conjecture 4.7 [8] posits the existence of the equivariant bridge $\varphi\colon \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$. Q9 [7] proves the bridge non-obstruction: the pathological Case 3 (obstruction to positivity) is excluded by the coercivity of the Mosco limit, leaving only the isotropic Case 1 ($A_H = A_Z$) and the anisotropic Case 2 ($A_H \neq A_Z$). Q10 [6] and U1 [9] then prove $A_H \to 2 = A_Z$ under the Q5a hypotheses and [U], establishing that only Case 1 survives in the limit. The bridge conjecture is thus resolved in the isotropic case conditionally on the Q5a hypotheses and [U]. Full bridge existence at finite q (Q7 Conjecture 4.7 in its original form) remains an open problem at the structural level; what is proved is the unique isotropic limit.

6.3 Structural significance: A_Z is not a free parameter

A central feature of the result is that A_Z is not a tunable constant of the effective geometry. It is fixed by the Casimir of the unique irreducible $\mathfrak{su}(2)$-module of dimension 3. This is consistent with the broader programme principle: physical parameters emerge from the spectral structure of the substrate, not from external inputs. The effective metric is therefore determined, up to the single temporal coefficient A_τ (subject to Q5b-O3 [3]), by the representation-theoretic data already established in O23 [1], O28–O29 [2, 5], and Q7 [8].

7 Summary and Open Problems

7.1 What this paper establishes

1. The sub-principal k_Z^2-term in $\sigma_2(L_{\mathrm{eff}})$ has a unique algebraic source: the Heisenberg commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$ (Proposition 3.1).
2. Under the equivariant bridge, the coefficient of this term is fixed by the $\mathfrak{su}(2)$-Casimir: $A_Z = C_{\mathfrak{su}(2)} = 2$ (Lemma 4.1, Proposition 4.2).
3. Consequently the effective spatial co-metric is isotropic with all eigenvalues equal to 2, and the full co-metric $g^{\mu\nu}$ is non-degenerate rank-4 Lorentzian (Theorem 1.1, Corollary 6.1).
4. Open problem Q5b-O2 of [3] is resolved under the hypotheses of Theorem 1.1.

7.2 Remaining open problems

Q8-O1 *(Bridge existence at finite q).* Q9 [7] proves the bridge non-obstruction (Case 3 excluded), and Q10–U1 close the isotropic limit $A_H = A_Z = 2$. What remains open is the finite-q structural question: does an explicit $\mathfrak{su}(2)$-equivariant isomorphism $\varphi_q\colon \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$ exist at each prime q, not just in the asymptotic limit? This is now a secondary open problem: the programme consequences are established without it.

Q8-O2 *(Unconditional isotropy — closed).* Q7 Conjecture 6.6 (asymptotic $A_H \to 2$ as $q \to \infty$) is now proved in Q10 [6] (structural argument under [U]) and U1 [9] (proof of [U] with rate $O(q^{-1/2})$), both conditionally on the Q5a hypotheses. This open problem is closed.

Q8-O3 *(Identification of A_τ and completion of Q5b-O3).* With $A_Z = 2$ established, the remaining free coefficient is A_τ (the temporal sector of the effective metric). Its identification from the Born–Infeld saturation data is the subject of Q5b-O3 [3].

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